

Zheng Rong JIA

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ORCID | GitHub | Google Scholar

RESEARCH INTERESTS

Medical AI, Deep Learning for Healthcare, Predictive Modeling, Multimodal Fusion, Time-Series Forecasting

EDUCATION

Macau University of Science and Technology
Bachelor of Science in Software Engineering

Macau, China
Graduated: Aug 2025

PUBLICATIONS

(*Corresponding author)

[2] **Zheng Rong JIA***, Kwong-Cheong Wong*. "DualTower-FT with an Adaptive Runtime Safeguard: A Deep Tabular Approach for ICU Stroke Mortality." *The 23rd Pacific Rim International Conference on Artificial Intelligence (PRICAI 2026)*, Guangzhou, China, Nov 17–20, 2026. Short paper, accepted. (CORE Rank B)

[1] **Zheng Rong JIA***, Kwong-Cheong Wong*. "Deep Learning for Stroke Mortality Prediction in eICU: A Dual-Tower Transformer Framework." *2026 IEEE 6th International Conference on Computer Communication and Artificial Intelligence (CCAI)*, Nanjing, China, May 22–24, 2026, pp. 76–81. DOI: [10.1109/CCAI69603.2026.11641950](https://doi.org/10.1109/CCAI69603.2026.11641950).

Awards: Best Industrial Paper; Best Presentation.

RESEARCH EXPERIENCE

Medical AI & Predictive Modeling

2025 – 2026

Lead Researcher

- Engineered a Dual-Tower Transformer (DT-Transformer) for stroke mortality prediction using the multi-center eICU database, narrowing the performance gap between deep learning architectures and gradient boosting baselines.
- Designed a decoupled architecture with a Self-Attention mechanism to independently process categorical demographics and numerical vitals, yielding a 14.4% relative AUPRC gain over the strongest neural baseline.
- Implemented an Adaptive Runtime Safeguard to ensure inference stability against physiological outliers, achieving a stable mean AUPRC of 0.6171 ± 0.0058 over five random seeds and a $3.4\times$ reduction in cross-seed variance versus a standard Transformer.

Macau University of Science and Technology

Sep 2024 – May 2025

Undergraduate Researcher (Final Year Project)

- Designed a multi-temporal scale LSTM model for daily, weekly, and quarterly power load forecasting.

- Achieved a 1.42% MAPE in daily predictions by applying dynamic outlier detection and feature selection (Pearson correlation and mutual information).
- Developed an interactive web-based decision support system for data visualization, outperforming traditional ARIMA models by 82% in quarterly accuracy.

SELECTED PROJECTS

Dual-Tower Transformer for eICU Stroke Mortality 2026

- Open-sourced the reference implementation of the DT-Transformer architecture and evaluation pipeline supporting the CCAI 2026 paper.

Data Preprocessing for eICU 2026

- Released a reproducible cohort extraction and feature engineering pipeline for the multi-center eICU database.

Academic Portfolio Website (zr-jia.github.io) 2026

- Developed and deployed a responsive personal academic website using GitHub Pages to aggregate research publications, CV, and open-source contributions.

PROFESSIONAL EXPERIENCE

Asia AI Education and Future Technology Association (AAE) Remote / Hong Kong, China *Young Ambassador and Researcher* Aug 2025 – Present

- Collaborated closely with Head Advisor Dr. Kwong-Cheong Wong to conduct research on Deep Learning for Stroke Mortality Prediction using the eICU database.
- Represented the association at IEEE CCAI 2026 (Nanjing, China), where the resulting paper received the Best Industrial Paper and Best Presentation awards.

China Southern Power Grid Guangzhou, China *Machine Learning Engineering Intern* Jul 2024 – Aug 2024

- Processed 60GB of multi-province power equipment inspection data to identify equipment fault frequencies and optimize inspection robot algorithms.
- Contributed to the development of closing and splitting functions for power inspection robots to support real-world operational accuracy.

ACADEMIC SERVICE

Pacific Rim International Conference on Artificial Intelligence (PRICAI) 2026 *Reviewer and Program Committee Member*

- Reviewed four submissions and was subsequently invited to serve as a Program Committee member.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript, MATLAB
- **Machine Learning & Data Science:** PyTorch, Pandas, NumPy, LSTM, Transformer, XGBoost, GRU, ARIMA
- **Tools & Frameworks:** Linux, Git, SQL, LaTeX, HTML/CSS